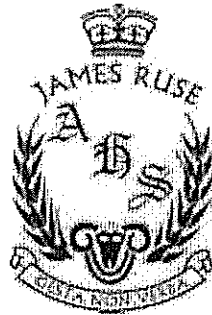


Name:	
Class:	



Year 12
Assessment task 2
Term 1 2017

MATHEMATICS EXTENSION 1

Time Allowed - 120 minutes

(plus 5 minutes Reading Time)

General Instruction:

- All questions may be attempted
- All questions are equal value
- Reference Sheet will be supplied
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or badly arranged work

Questions 1-5 are to be completed on the multiple choice sheet, then Questions 6-12 to be completed on separate sheets of paper and handed in in separate bundles. Questions must show your Candidate Number (in the top right hand corner).

SECTION I
MULTIPLE CHOICE QUESTIONS

Attempt Questions 1 – 5 (1 mark each).

Allow approximately 8 minutes for this section

1. Find the fifth term in the expansion of $\left(3x^2 - \frac{1}{2x}\right)^8$.

- A) $\binom{8}{4} \left(\frac{3}{2}\right)^4 x^4$;
- B) $\binom{4}{8} \left(\frac{2}{3}\right)^4 x^{-4}$;
- C) $\binom{8}{4} \left(\frac{3}{2}\right)^4 x^{-4}$;
- D) $\binom{4}{8} \left(\frac{2}{3}\right)^4 x^4$;

2. By using substitution $u = \tan^{-1}x$ evaluate $\int \frac{e^{\tan^{-1}x}}{1+x^2} dx$

- A) $e^{x^2} + c$;
- B) $e^{\tan^{-1}x} + c$;
- C) $e^x + c$;
- D) $e^{1+x^2} + c$.

3. Which one of these functions has an inverse relation that is not a function?

- A) $y = x^3$;
- B) $y = \ln x$;
- C) $y = \sqrt{x}$;
- D) $y = |x|$.

4. If $\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$, then $\sin^{-1}(\sin x)$ is equal to:

- A) $x - 2\pi$
- B) $\pi + x$
- C) $\frac{\pi}{2} - x$
- D) $\pi - x$

5. Evaluate $\lim_{n \rightarrow -\infty} \frac{(n^4 + 3n^2) \tan^{-1}(2^n)}{n^4 - 1}$.

- A) 1;
- B) $\frac{\pi}{2}$;
- C) $-\frac{\pi}{2}$;
- D) 0.

END OF SECTION I

SECTION II
EXTENDED RESPONSE (77 marks)

77 marks

Attempt Questions 6 – 12

Allow approximately 1 hour and 52 minutes for this section

MARKS:

QUESTION 6. (11 marks)**COMMENCE A NEW PAGE**

(a) Evaluate

$$\lim_{x \rightarrow 0} \left(\frac{\sin 4x}{5x} + \cos x \right).$$

2

(b) Use the binomial theorem to find the term independent of x in the expansion of

$$\left(\frac{4}{x} + x^2 \right)^{15}$$

2

(c) Prove by mathematical induction that $n^3 + 2n$ is divisible by 3, for all positive integers n .

3

(d) Sketch the graph of the function $f(x) = 3 \cos^{-1}(2x - 1)$.

Clearly indicate the domain and range of the function.

2

(d) A spherical balloon is expanding so that its volume, $V \text{ m}^3$, increases at a constant rate of 72 m^3 per second. What is the rate of increase of the surface area when the radius is 12 metres? You may use the formula $V = \frac{4}{3}\pi r^3$ for the volume of a sphere and $S = 4\pi r^2$ for its surface area.

2

QUESTION 7. (11 marks)**COMMENCE A NEW PAGE**

(a) If

$$\int \frac{\ln x}{x\sqrt{1 + \ln x}} dx = F(x) + C,$$

using substitution $u = 1 + \ln x$ (i) find $F(x)$,

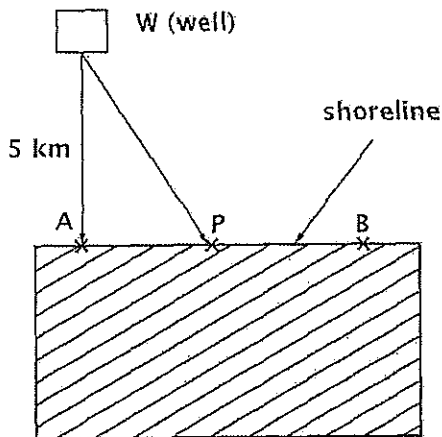
2

(ii) hence, find $F(e^3) - F(1)$

2

MARKS:

- (b) An offshore oil well is located at a point W , which is 5 km from the closest shorepoint, A , on a straight shoreline. The oil is to be piped to a shorepoint B that is 8 km from A by piping it on a straight line underwater from W to some shorepoint, P , between A and B and then on to B via a pipe along the shoreline.



The cost of laying the pipe is \$125 000 per km underwater and \$75 000 per km over land. Let $x\text{ km}$ be the distance between A and P , and C (in thousands of dollars) be the cost of the entire pipeline.

- (i) Show that the cost is given by $C = 125\sqrt{x^2 + 25} + 75(8 - x)$ 2
- (ii) Find the domain of x . 1
- (iii) Find where the point P should be located to minimise the cost of laying the pipe, then find this minimised cost. 4

QUESTION 8. (11 marks)

COMMENCE A NEW PAGE

- (a) Use mathematical induction to prove that, for all integers n with $n \geq 1$

$$3 \cdot 2! + 7 \cdot 3! + 13 \cdot 4! + \cdots + (n^2 + n + 1)(n + 1)! = n(n + 2)! \quad 4$$

- (b) Find the angle that $y = \tan^{-1}(2x)$ makes with the y -axis at the origin.
Give your answer to the nearest degree. 3

- (c) For a given function

$$f(x) = x^2 - \left(m - \frac{2}{3}\right)x^3 - mx + 1,$$

find the real number, m , for which the function does not have a minimum or maximum? 4

MARKS:

QUESTION 9. (11 marks)

COMMENCE A NEW PAGE

- (a) Let
- $x = 3 \sin \theta$
- and find the following integral

$$\int \frac{x^2}{\sqrt{9-x^2}} dx \quad 4$$

- (b) Two points
- $P(2ap, ap^2)$
- and
- $Q(2aq, aq^2)$
- lie on the parabola
- $x^2 = 4ay$
- .

- (i) Show that the equation of the tangent to the parabola at
- P
- is
- $y = px - ap^2$
- .
- 2

- (ii) The tangent at
- P
- and the line through
- Q
- parallel to the
- y
- axis intersect at
- T
- .
-
- Find the coordinates for
- T
- .
- 2

- (iii) Write down the coordinates of
- M
- , the midpoint of
- PT
- .
- 1

- (iv) Following information given above, draw corresponding diagram, showing
-
- coordinates of all points.
- 1

- (iv) Determine the locus of
- M
- when
- $pq = -1$
- 1

QUESTION 10. (11 marks)

COMMENCE A NEW PAGE

- (a) Let
- $u = \sin x$
- , find the following indefinite integral

$$\int \sin^3 x \cos x dx \quad 2$$

- (b) Isabella invests \$P at 8% per annum compounded annually. She intends to
-
- withdraw \$3000 at the end of each of the next six years, immediately after the
-
- interest is paid, to cover school fees.

- (i) Write down an expression for the amount \$A1 remaining in the account
-
- following the withdrawal of the first \$3000.
- 1

- (ii) Find an expression for the amount \$A2 remaining in the account after
-
- the second withdrawal.
- 1

- (iii) Calculate the amount \$P that Isabella needs to invest so that she is just
-
- able to cover the school fees.
- 3

- (c) Prove by induction that

$$\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^n} = \frac{1}{x^n(x-1)}$$

for all positive integers $n \geq 1$, $x \neq 0, 1$. 4

MARKS:

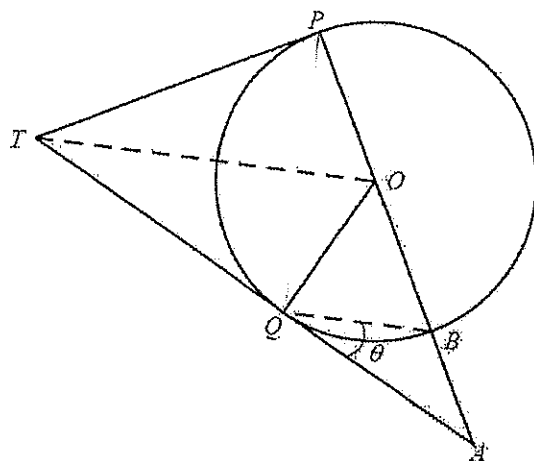
QUESTION 11. (11 marks)

COMMENCE A NEW PAGE

- (a) Find the area bounded by the curves $y = \cos x$ and $y = \sin(2x)$ and the x -axis for $0 \leq x \leq \frac{\pi}{2}$

3

(b)



From an external point T , tangents are drawn to a circle with centre O , touching the circle at P and Q . Angle PTQ is acute. The diameter PB produced meets the tangent TQ at A . Let $\angle AQB = \theta$.

- (i) Copy the diagram above into your answer booklet.
 (ii) Prove that $\angle PTQ = 2\theta$.
 (iii) Prove that $\triangle PBQ$ and $\triangle TOQ$ are similar.
 (iv) Hence show that $BQ \times OT = 2 \times OP^2$.

3

3

2

QUESTION 12. (11 marks)

COMMENCE A NEW PAGE

- (a) For the binomial expansion

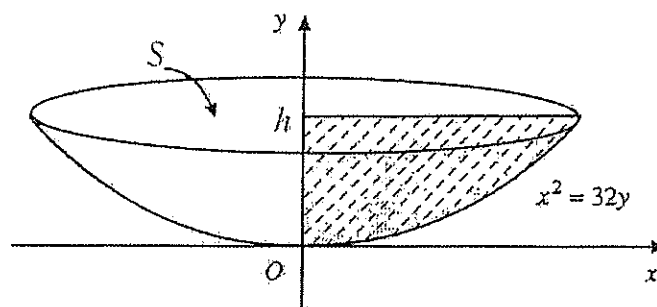
$$\left(\frac{a\sqrt[3]{a}}{b} + \frac{1}{\sqrt[15]{a^{28}}} \right)^n, \quad n \in \mathbb{N}, \quad a, b \neq 0, \quad a, b \text{ are variables}$$

the sum of the coefficients of the first three terms is 79.
 Find the term independent of a .

4

MARKS:

(b)



- (i) The part of the curve $x^2 = 32y$ between $y = 0$ and $y = h$ is rotated about the y axis. Show that the volume enclosed is given by $V = 16\pi h^2$.

1

- (ii) The diagram represents the water in a dam on a farm. The depth of the water is h metres, the volume of the water in the dam is $V \text{ m}^3$, and the area of the surface of the water is $S \text{ m}^2$. The water in the dam evaporates according to the rule $\frac{dV}{dt} = -kS$, where k is a positive constant, and t is the time in hours. Show that $\frac{dh}{dt} = -k$.

3

- (iii) Initially the dam contains $64\pi \text{ m}^3$ of water. Calculate how long it will take for the dam to empty by evaporation when $k = 0.001$.

3

Extension 1 Mathematics

1.

$$T_{k+1} = \binom{n}{k} a^{n-k} (b)^k$$

$$T_5 = \binom{8}{4} (3)^4 \left(\frac{1}{2x}\right)^4$$

$$= \binom{8}{4} \left(\frac{3}{2}\right)^4 x^{-4}$$

part C.

2. $u = \tan^{-1} x$

$$\frac{du}{dx} = \frac{1}{1+x^2}$$

$$\therefore \int e^u du.$$

$$= e^u + c$$

$$= e^{\tan^{-1} x} + c.$$

part B

3 part D

4. $\sin^{-1}(\sin x)$

x is in 2nd and
third quadrant.

$$\sin^{-1}\left(\sin \frac{7\pi}{6}\right) = \frac{7\pi}{6}$$

$$\therefore \pi - x.$$

$$\sin^{-1}\left(\sin \frac{5\pi}{6}\right) = \frac{\pi}{6}$$

$$\underline{\pi - x.}$$

$\therefore$ part D.

5. $\lim$

$$n \rightarrow -\infty.$$

$$\frac{(1+\theta) \tan^{-1} \frac{1}{\theta}}{1-\theta}$$

$$= \frac{(1+\theta) \tan^{-1} 0}{1-\theta}$$

$$= \underline{0}$$

part D

1/2

MATHEMATICS Extension 1 : Question 7

Suggested Solutions

Marks

Marker's Comments

(a)(i) $\int \frac{\ln x \, dx}{x\sqrt{1+\ln x}}$ using $u = 1 + \ln x$
 $\frac{du}{dx} = \frac{1}{x}$
 $du = \frac{dx}{x}$
 $= \int \frac{(u-1) du}{\sqrt{u}}$
 $= \int (\sqrt{u} - u^{-1/2}) du$
 $= \frac{2}{3} u^{3/2} - 2u^{1/2} + C$
 $= \frac{2}{3} (1+\ln x)^{3/2} - 2(1+\ln x)^{1/2} + C$
 or $\frac{2}{3} (1+\ln x)^{1/2} (\ln x - 2)$

①

①

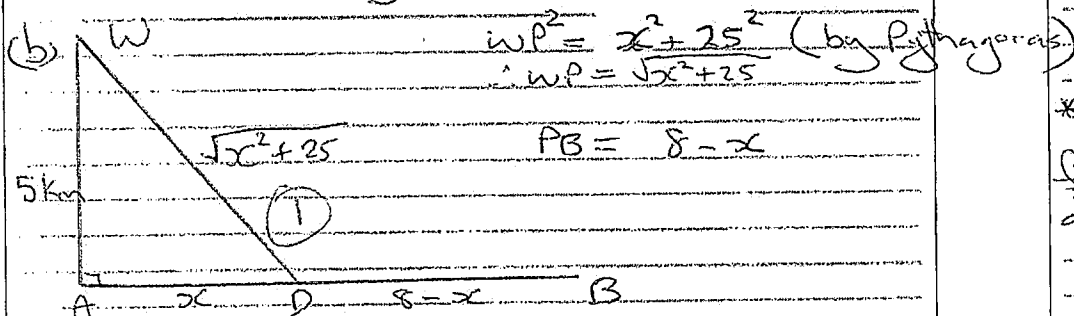
A lot of students forgot to substitute back.

(ii) $F(e^3) - F(1)$

$F(e^3) - F(1) = \frac{2}{3} (1+\ln e^3)^{3/2} - 2(1+\ln e^3)^{1/2} - \frac{2}{3} (1+\ln 1)^{3/2} + 2(1+\ln 1)^{1/2}$
 $= \frac{2}{3} (4)^{3/2} - 2(4)^{1/2} - \frac{2}{3} (1)^{3/2} + 2(1)^{1/2}$
 $= \frac{16}{3} - 4 - \frac{2}{3} + 2$
 $= \frac{8}{3}$

①

* If students didn't sub back in (i) then they made the question easier for (b) so they only got one mark.



①

Cost = $125000 \sqrt{x^2 + 25} + 75000(8 - x)$

$\therefore C = 125 \sqrt{x^2 + 25} + 75(8 - x)$

(ii) $x > 0$ as it's a distance
 $8 - x > 0$
 $\therefore x < 8$

$\therefore 0 < x < 8$ or $0 \leq x \leq 8$

①

(iii) $C = 125(x^2 + 25)^{1/2} + 75(8 - x)$

$\frac{dC}{dx} = 125 \times \frac{1}{2} (x^2 + 25)^{-1/2} \times 2x + -75$

$0 < x < 8$ scored no marks.

2/2

MATHEMATICS Extension 1 : Question 7 continued

Suggested Solutions

Marks

Marker's Comments

$$\therefore \frac{dc}{dx} = \frac{125x}{\sqrt{x^2+25}} - 75$$

①

stat. points when $\frac{dc}{dx} = 0$

$$\therefore \frac{125x}{\sqrt{x^2+25}} = 75$$

$$\frac{5x}{\sqrt{x^2+25}} = 3$$

$$25x^2 = 9(x^2+25)$$

$$16x^2 = 225$$

$$x = \pm \frac{15}{4}$$

$$x = \frac{15}{4} \quad (\text{as } 0 < x \leq 8)$$

①

x	3	$\frac{15}{4}$	4
$\frac{dc}{dx}$	-10.69	0	3.09
grad.	\	=	/

* needed to test properly.

$\therefore$ rel. min at $x = \frac{15}{4}$

①

∴ minimum cost is 0

* A lot of students forgot to actually find the minimum cost. It is

$$C = 125 \sqrt{\left(\frac{15}{4}\right)^2 + 25} + 75 \left(8 - \frac{15}{4}\right)$$

$$= 125 \times \frac{25}{4} + 75 \times \frac{17}{4}$$

$$= 1100 \text{ thousands}$$

∴ minimum cost is \$1100 000

①

Suggested Solutions

Marks Awarded

Marker's Comments

8a) Step 1 Prove true for $n=1$

$$\begin{aligned} \text{LHS} &= (1+1+1)(1+1)! \\ &= 3 \times 2! \\ &= 6 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= n(n+2)! \\ &= 1(1+2)! \\ &= 3! \\ &= 6 \end{aligned}$$

LHS = RHS true for $n=1$

Need to show LHS & RHS separately then say LHS = RHS & true for $n=1$ to gain 1st mark.

Step 2

Assume true for $n=k$

$$3 \cdot 2! + 7 \cdot 3! + 13 \cdot 4! + \dots + (k^2 + k + 1)(k+1)! = k(k+2)!$$

Step 3

Prove true for $n=k+1$

$n \in \mathbb{Z}$

RTP

$$\begin{aligned} 3 \cdot 2! + 7 \cdot 3! + 13 \cdot 4! + \dots + ((k+1)^2 + (k+1)+1)(k+1+1)! \\ \downarrow \\ (k+1)^2 + (k+2)(k+2)! = (k+1)(k+3)! \end{aligned}$$

LHS

$$\begin{aligned} 3 \cdot 2! + 7 \cdot 3! + 13 \cdot 4! + \dots + (k^2 + k + 1)(k+1)! + (k+1)^2 + (k+2)(k+2)! \\ \underbrace{\hspace{10em}}_{\text{by assumption}} \\ k(k+2)! + [(k^2 + 2k + 1) + (k+2)](k+2)! \\ k(k+2)! + (k^2 + 3k + 3)(k+2)! \end{aligned}$$

1 mark awarded to state by assumption

Suggested Solutions

Marks Awarded

Marker's Comments

$$\begin{aligned}
 & \frac{k \times 3}{k \times 1} \left\{ \begin{aligned} & (k+2)! (k+k^2+3k+3) \\ & (k+2)! (k^3+4k+3) \\ & (k+2)! (k+3)(k+1) \end{aligned} \right\}
 \end{aligned}$$

①

tidy up

$$* (k+3)(k+2)! = (k+3)!$$

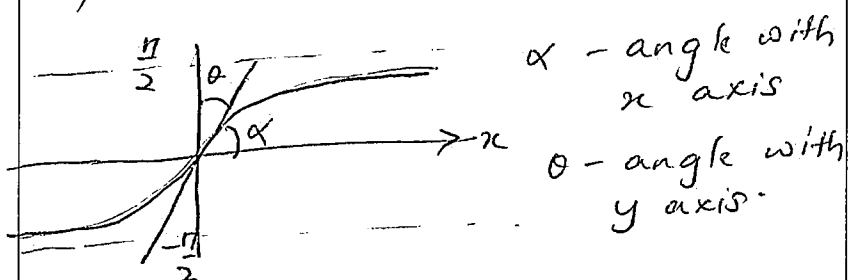
$$\therefore (k+3)! (k+1)! = \text{RHS}$$

①

to prove = RHS

Hence if true for $n=k$, then
 true for $n=k+1$. $\therefore$ Result
 true by mathematical induction
 for all integers $n \geq 1$

$$b) y = \tan^{-1} 2x$$



$$\tan y = 2x$$

$$x = \frac{1}{2} \tan y$$

$$\begin{aligned}
 \frac{dx}{dy} &= \frac{1}{2} \sec^2 y \\
 &= \frac{1}{2} \left(\frac{1}{\cos^2 y} \right)
 \end{aligned}$$

①

 able to
differentiate

For $\frac{dy}{dx} = 2\cos^2 y$ at 0

$$\begin{aligned} m &= 2\cos^2(0) \\ &= 2(1) \\ &= 2 \end{aligned}$$

$\therefore \tan \alpha = 2$

$$\begin{aligned} \alpha &= \tan^{-1}(2) \\ &= 63.43^\circ \text{ or } 63^\circ 26' \end{aligned}$$

— (1)

$\therefore$ angle with y axis is

$$\begin{aligned} &90 - 63^\circ 26' \\ &= 27^\circ \text{ (nearest degree)} \end{aligned}$$

— (1)

OR.

$$\frac{dy}{dx} = \frac{1}{1+(2x)^2} \times 2.$$

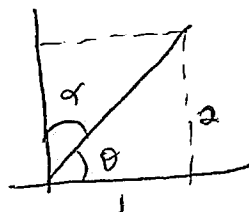
$$= \frac{2}{1+4x^2} \quad \text{when } x=0$$

$$\tan \theta = \frac{2}{1}$$

$$\theta = 63^\circ$$

$$\alpha = 90 - 63$$

$$= 27^\circ$$



Suggested Solutions

Marks Awarded

Marker's Comments

$$c) f(x) = x^2 - (m - \frac{2}{3})x^3 - mx + 1$$

$$f'(x) = 2x - 3x^2(m - \frac{2}{3}) - m$$

$$= -3(m - \frac{2}{3})x^2 + 2x - m$$

$$a = -3(m - \frac{2}{3})$$

$$b = 2$$

$$c = -m$$

$$\Delta = b^2 - 4ac$$

$$= (2)^2 - 4(-3(m - \frac{2}{3}))(-m)$$

$$= 4 - 4(3m^2 - 2m)$$

$$= 4 - 12m^2 + 8m$$

To not have min or max ie no roots

$$\Delta < 0 \quad \therefore f'(x) \neq 0$$

$$4 - 12m^2 + 8m < 0$$

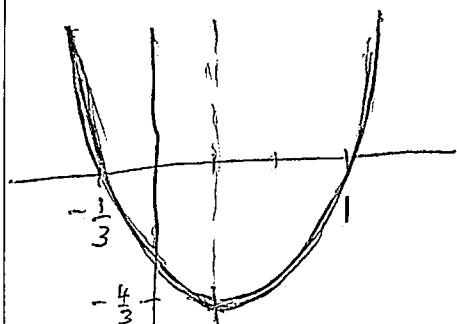
$$4(1 - 3m^2 + 2m) < 0$$

$$-(3m^2 - 2m - 1) < 0$$

$$3m^2 - 2m - 1 > 0$$

$$(3m+1)(m-1) > 0$$

$$m = -\frac{1}{3}, 1$$



$$\therefore m > 1$$

$$\text{or } m < -\frac{1}{3}$$

know that
have to differentiate

if haven't
mentioned $\Delta < 0$
no marks awarded

cannot write
inequations of
 $m \geq 1$ $m \leq -\frac{1}{3}$
coz it doesn't

MATHEMATICS Extension 1 : Question 9

Suggested Solutions

Marks Awarded

Marker's Comments

$$a) \int \frac{x^2}{\sqrt{9-x^2}} dx = \int \frac{9 \sin^2 \theta}{\sqrt{9-9 \sin^2 \theta}} \cdot dx$$

4 marks

let $x = 3 \sin \theta$
 $\frac{dx}{d\theta} = 3 \cos \theta$
 $\therefore dx = 3 \cos \theta \cdot d\theta$

Replace dx with $3 \cos \theta \cdot d\theta$
 and simplify denominator

$$= \int \frac{9 \sin^2 \theta}{3 \sqrt{1-\sin^2 \theta}} \cdot 3 \cos \theta \cdot d\theta$$

$$= \int \frac{9 \sin^2 \theta \times 3 \cos \theta \cdot d\theta}{3 \cos \theta}$$

$$= \int 9 \sin^2 \theta \cdot d\theta$$

We know $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$

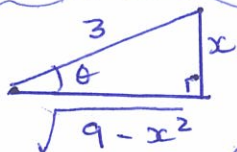
$$= \frac{9}{2} \int (1 - \cos 2\theta) \cdot d\theta$$

$$= \frac{9}{2} \left[\theta - \frac{\sin 2\theta}{2} \right] + C$$

$$= \frac{9\theta}{2} - \frac{9 \sin 2\theta}{4} + C$$

We know

$$\sin \theta = \frac{x}{3}$$



$$\therefore \cos \theta = \frac{\sqrt{9-x^2}}{3}$$

$$= \frac{9\theta}{2} - \frac{2 \cos \theta \sin \theta \times 9}{4} + C$$

as $\sin 2\theta = 2 \cos \theta \sin \theta$

and

$$\theta = \sin^{-1} \left(\frac{x}{3} \right)$$

$$= \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) - 2 \times \frac{\sqrt{9-x^2}}{3} \times \frac{x}{3} \times \frac{9}{4} + C$$

$$= \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) - \frac{x \sqrt{9-x^2}}{2} + C$$

Also accepted $= \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) - \frac{3x}{2} \sqrt{\frac{1-x^2}{9}} + C$

✓ 1 mark for replacing dx with $3 \cos \theta \cdot d\theta$

✓ 1 mark for simplifying to $\frac{9}{2} \int (1 - \cos 2\theta) \cdot d\theta$

✓ 1 mark for integrating to $\frac{9\theta}{2} - \frac{9 \sin 2\theta}{4} + C$

✓ Simplifying to answer given

MATHEMATICS Extension 1 : Question.....9b

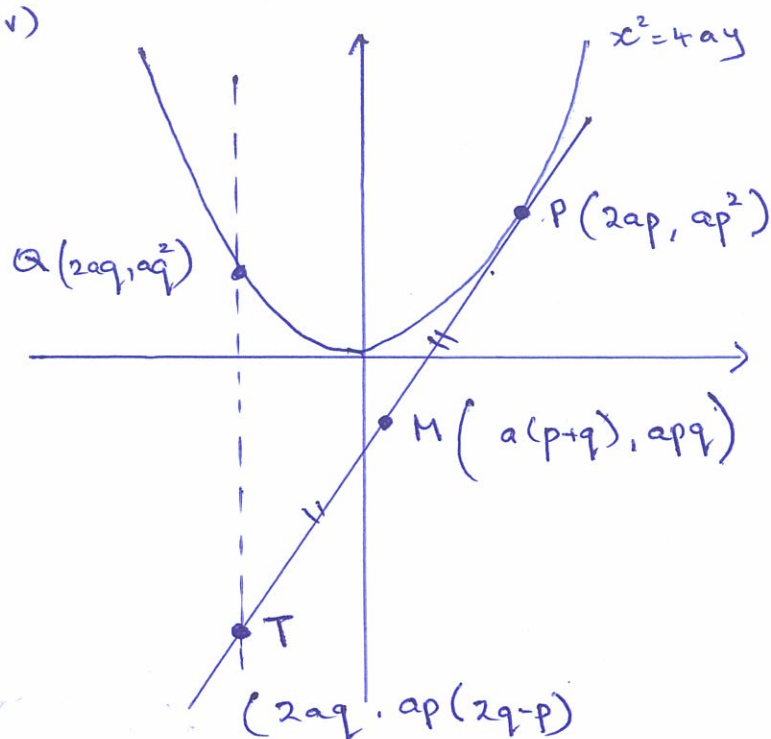
Suggested Solutions	Marks Awarded	Marker's Comments
b(i) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ $x^2 = 4ay \Rightarrow \therefore y = \frac{x^2}{4a}$ and $\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$ at point P $x = 2ap$ $\therefore \frac{dy}{dx} = \frac{2ap}{2a} = p$ Equation : $y - y_1 = m(x - x_1)$ $y - ap^2 = p(x - 2ap)$ $y - ap^2 = px - 2ap^2$ $\therefore y = px - ap^2$	2 marks	well done ✓ 1 mark for getting gradient p ✓ 1 mark for using equation formula correctly.
b(ii) Substitute $x = 2aq$ into equation $\therefore y = px - ap^2$ $= p(2aq) - ap^2$ $= 2apq - ap^2$ $\therefore$ Coordinates of T is $(2aq, 2apq - ap^2)$ or $(2aq, ap(2q - p))$	2 marks	As T is on parallel line from Q the x value is the same as 2aq ✓ 1 mark for recognising $x = 2aq$ ✓ 1 mark for getting y value
b(iii). Midpoint of PT. $P(2ap, ap^2)$ and $T(2aq, ap(2q - p))$ $\left(\frac{2ap + 2aq}{2}, \frac{ap^2 + 2apq - ap^2}{2} \right)$ $= (ap + aq, apq)$ $= (a(p + q), apq)$	1 mark.	✓ 1 mark for getting both x and y values correctly.

Suggested Solutions

Marks Awarded

Marker's Comments

iv)



1 mark.

All coordinates of P, Q, M and T had to be written

Few students forgot to label M and lost a mark.

 v) For M $(ap+aq, apq)$

$$x = a(p+q) \text{ and } y = apq$$

$$y = a(-1)$$

$$y = -a$$

Since the y coordinate of M is $-a$ and which is independent of the variables p and q then the locus M is the line $y = -a$.

This is also the directrix of the parabola.

$$x^2 = 4ay$$

1 mark

One had to have $y = -a$ as the answer.

Some had $y = -a$ in their working out but used that to do other calculations. They used $y = -a$ but did not conclude $y = -a$ is locus. No marks awarded

Task 2 : 2017

MATHEMATICS Extension 1 : Question.....

10

JRHS

Suggested Solutions

Marks

Marker's Comments

10a) $I = \int \sin^3 x (\cos x dx)$

$u = \sin x$

$du = \cos x dx$

$$\therefore I = \int u^3 du = \frac{u^4}{4} + C$$

$$= \frac{\sin^4 x}{4} + C$$

①

integrand in u

①

final correct answer in terms of x .

b) (i) $A_1 = (1.08)' P - 3000$

(ii) $A_2 = (1.08') A_1 - 3000$

$$= 1.08 (1.08 P - 3000) - 3000$$

$$= 1.08^2 P - 3000 (1 + 1.08)$$

①

For correct expression

①

For correct expression

(iii) $A_3 = 1.08^2 P - 3000 (1 + 1.08 + 1.08^2)$

$\vdots$

$$A_6 = 1.08^6 P - 3000 (1 + 1.08 + \dots + 1.08^5)$$

$$= 1.08^6 P - 3000 \times \left[\frac{1(1.08^6 - 1)}{1.08 - 1} \right]$$

$$= 1.08^6 P - 37500 (1.08^6 - 1)$$

①

For A_6 or A_n

①

GP with $a=1, r=1.08$
 $n=6$

for S_n for GP

But $A_6 = 0$

$$\Rightarrow 1.08^6 P = 37500 (1.08^6 - 1)$$

i.e $P = \frac{37500 (1.08^6 - 1)}{1.08^6}$

$$= \$13\,868.64 \text{ (nearest cent)}$$

①

Final investment amount

* If $n=5$, max 4

* If $A_6=18000$, max 4

* If regular investments, max 3

Task 2 : 2017

MATHEMATICS Extension 1 : Question 10

TRAHS

Suggested Solutions

Marks

Marker's Comments

c) Let the proposition be $P(n)$

i.e. $P(n) : \frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^n} = \frac{1}{x^n(x-1)}$

Prove $P(n)$ true for $n=1$

$$\text{LHS (of } P(n)) = \frac{1}{x-1} - \frac{1}{x} = \frac{1}{x(x-1)}$$

$$\text{RHS (of } P(n)) = \frac{1}{x^1(x-1)} = \text{LHS}$$

$\therefore P(n)$ is true for $n=1$

Assume $P(n)$ is true when $n=k \in \mathbb{Z}^+$

i.e. $\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^k} = \frac{1}{x^k(x-1)}$

Prove $P(n)$ true when $n=k+1$

i.e. To prove:

$$\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^k} - \frac{1}{x^{k+1}} = \frac{1}{x^{k+1}(x-1)}$$

Now

$$\text{LHS} = \left(\frac{1}{x-1} - \frac{1}{x} - \frac{1}{x^2} - \dots - \frac{1}{x^k} \right) - \frac{1}{x^{k+1}}$$

$$= \frac{1}{x^k(x-1)} - \frac{1}{x^{k+1}}$$

$$= \frac{x - (x-1)}{x^{k+1}(x-1)} = \frac{1}{x^{k+1}(x-1)} = \text{RHS}$$

$\therefore P(n)$ is true when $n=k+1$

Hence by the principle of Mathematical Induction $P(n)$ is true $\forall n \geq 1$

1

For base case

1

Equation for Inductive step for $n=k+1$

1

Correct algebra in proving LHS = RHS

1

For overall structure including appropriate conclusion

* If $n=2 \dots$ max 3

* If $x=2 \dots$ max 2

* If $\frac{1}{x^k}$ equation for inductive step - max.

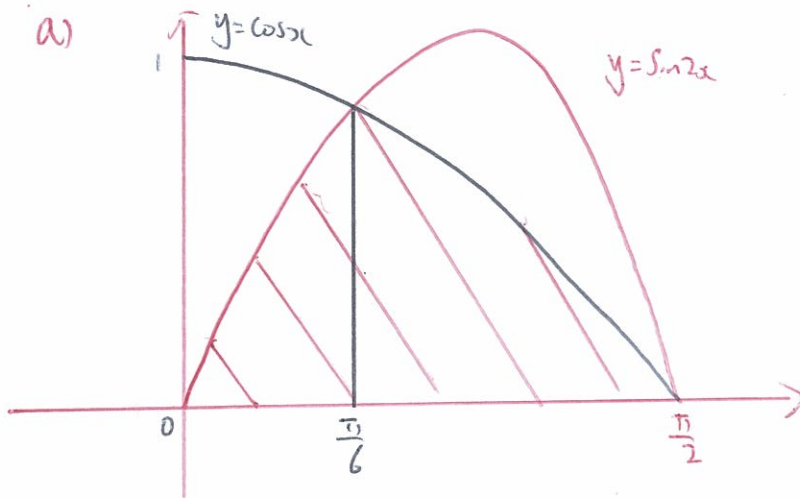
* If $\frac{1}{x^k}$ missing from inductive step

Suggested Solutions

Marks

Marker's Comments

11. a)


 Solve $\cos x = \sin 2x$

$$\cos x = 2 \sin x \cos x$$

$$2 \sin x \cos x - \cos x = 0$$

$$\cos x (2 \sin x - 1) = 0 \rightarrow \cos x = 0 \text{ or } \sin x = \frac{1}{2}$$

$$\therefore x = \frac{\pi}{2} \text{ or } x = \frac{\pi}{6}$$

1

$$\therefore A = \int_0^{\frac{\pi}{6}} \sin 2x \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos x \, dx$$

1

$$= \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}} + \left[\sin x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= -\frac{1}{2} \left(\frac{1}{2} - 1 \right) + \left(1 - \frac{1}{2} \right)$$

$$= \frac{1}{4} + \frac{1}{2}$$

$$= \frac{3}{4} u^2$$

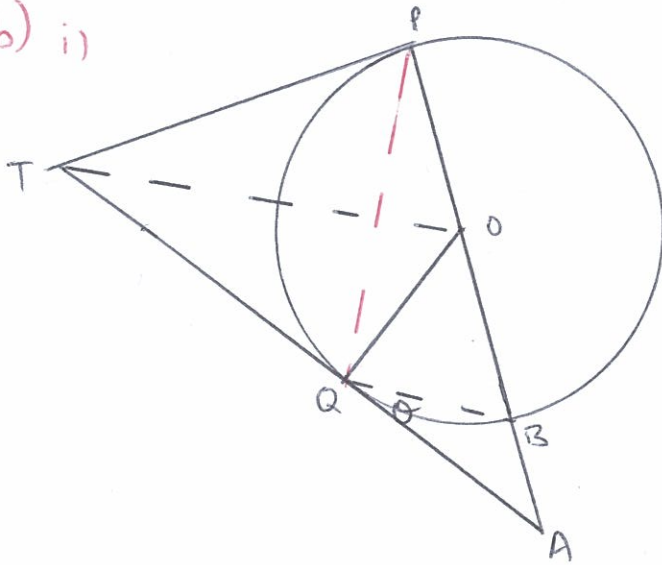
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Suggested Solutions

Marks

Marker's Comments

b) i)



ii) Construct PQ

$\angle QPB = \theta$ (angle between chord and tangent is equal to angle in the alternate segment) — (1)

$\angle TPO = 90^\circ$ (radius OP is perpendicular to tangent TP at point of contact).

$\angle TQO = 90^\circ$ (radius OQ is perpendicular to tangent TQ at the point of contact)

$\therefore \angle TPO + \angle TQO = 180^\circ$

$\therefore TPOQ$ is cyclic — (1)

$\angle QOB = 2\theta$ (angle at the centre is twice the angle at the circumference standing on the same arc)

$\therefore \angle PTQ = 2\theta$ (exterior angle of a cyclic quadrilateral is equal to the interior opposite angle) — (1)

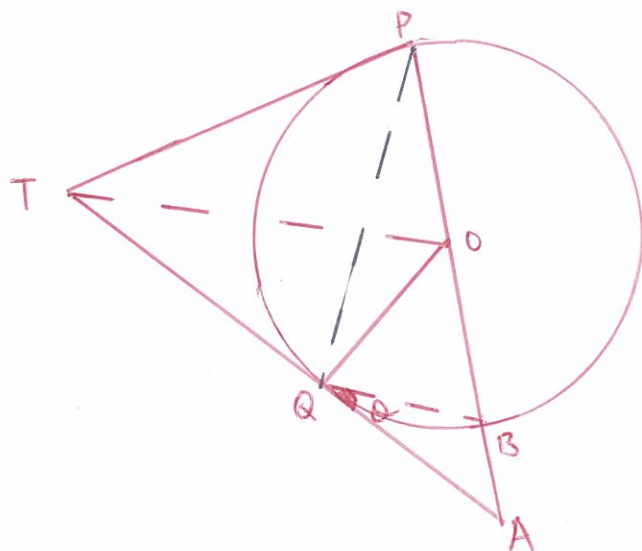
Suggested Solutions

Marks

Marker's Comments

b)

i)


 ii) In $\triangle PTO$ and $\triangle QTO$,

 $\angle OQT = 90^\circ$ (angle between radius OQ and tangent TQ)

 $\angle OPT = 90^\circ$ (angle between radius OP and tangent PT)

 $\therefore \angle OQT = \angle OPT = 90^\circ$
 TO is common

 $PO = QO$ (radii of a circle)

 $\therefore \triangle PTO \cong \triangle QTO$ (RHS)

 Construct PA ,

 $\angle QPB = \theta$ (angle between chord and tangent is equal to the angle in the alternate segment)

 $\therefore \angle QOB = 2\theta$ (angle at the centre is twice the angle at the circumference standing on the same arc)

 $\angle POQ = 180 - 2\theta$ (angle sum of a straight angle)

 $\angle POQ = \angle POT + \angle QOT$
 $= 2\angle QOT$ (corresponding angles, $\triangle POT \cong \triangle QOT$)

Suggested Solutions	Marks	Marker's Comments
$\therefore \angle QOT = 90 - \theta$ $\therefore \angle QTO = 180 - \angle OQT - \angle QOT \text{ (angle sum } \triangle TOQ)$ $= 180 - (90 - \theta) - 90^\circ$ $= \theta$ $\therefore \angle PTO = \theta \text{ (corresponding angles } \triangle PTO \cong \triangle QTO)$ $\therefore \angle PTQ = 2\theta.$		

Suggested Solutions	Marks	Marker's Comments
iii) $QO = BO$ (radii of a circle) $\therefore \angle OQB = \angle OBQ$ (equal angles are opposite to equal sides in $\triangle BOQ$) $\angle QOB + \angle QBO + \angle BQO = 180^\circ$ (angle sum of $\triangle BOQ$) $\therefore \angle QOB + 2\angle OBQ = 180^\circ$ (proven above) $\therefore \angle OBQ = \frac{180 - QOB}{2}$ $\angle QOB = 2\theta$ (proven above) $\therefore \angle OBQ = 90 - \theta$ $\therefore$ In $\triangle PBQ$ and $\triangle TOQ$, $\angle PQB = 90^\circ$ (angle in a semi-circle) $\angle OQT = 90^\circ$ (proven above) $\therefore \angle PQB = \angle OQT$ $\angle PBQ = 90 - \theta = \angle TOQ$ (proven above) $\therefore \triangle PBQ \parallel \triangle TOQ$ (equiangular) Using Equiangular: 1 - for proving each pair of angles equal 1 - for $\triangle PBQ \parallel \triangle TOQ$ (equiangular) *Students receive $1/3$ if any part of the proof is invalid		

Suggested Solutions	Marks	Marker's Comments
iv) $\frac{BQ}{QO} = \frac{PB}{TO}$ (corresponding sides are in the same ratio) $\triangle PBQ \parallel \triangle TOQ$ $\therefore BQ \times TO = PB \times QO$ $= 2 \times OP \times OQ$ (diameter is twice the radius in a circle) $= 2 \times OP^2$ (QO and OP are radii of the circle)	(i) (1)	

MATHEMATICS Extension 1 : Question...12..

Suggested Solutions

Marks

Marker's Comments

$$a) T_{r+1} = {}^nC_r \left(\frac{a\sqrt[3]{a}}{b} \right)^{n-r} \left(\frac{1}{15\sqrt{a^{28}}} \right)^r$$

$$= {}^nC_r \frac{a^{\frac{4}{3}(n-r) - \frac{28}{15}(r)}}{b^{n-r}}$$

$${}^nC_0 + {}^nC_1 + {}^nC_2 = 79$$

$$1 + \frac{n!}{(n-1)!} + \frac{n!}{2!(n-2)!} = 79$$

$$1 + n + \frac{1}{2}n(n-1) = 79$$

$$2 + 2n + n^2 - n = 158$$

$$n^2 + n - 156 = 0$$

$$(n+13)(n-12) = 0$$

$$n = 12, -13$$

$$\therefore n = 12, n \geq 0$$

for term independent of a :

$$\frac{4}{3}(n-r) - \frac{28}{15}(r) = 0$$

$$\frac{4}{3}n - \frac{4}{3}r - \frac{28}{15}r = 0$$

$$\frac{4}{3}n = \frac{16}{5}r$$

$$r = \frac{5}{12}n$$

$$\text{when } n = 12$$

$$r = 5$$

$\therefore$ Term independent of a is:

$$T_6 = {}^{12}C_5 b^{12-5}$$

$$= \frac{792}{b^7}$$

$$b) i) x^2 = 32y$$

$$V = \pi \int_0^h x^2 dy$$

$$= \pi \int_0^h 32y dy$$

$$= \pi [16y^2]_0^h$$

$$= \pi (16h^2 - 0)$$

$$= 16h^2\pi$$

1

if used 'guess and check' students lost this mark.

1

1

1

} 1

MATHEMATICS Extension 1 : Question 12.

Suggested Solutions

Marks

Marker's Comments

ii) $S = \pi r^2$

$x^2 = 32y$

$x = \text{radius}$ $y = \text{height}$

$\therefore r^2 = 32h$

$\therefore S = 32h\pi$

$\frac{dV}{dh} = 32h\pi$

$\therefore \frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$

$= \frac{1}{32h\pi} \times -kS$

$= \frac{1}{32h\pi} \times -k(32h\pi)$

$= -k$

iii) $\frac{dh}{dt} = -k$

$dh = -k dt$

$h = -kt + C$

when $t = 0$, $V = 64\pi$

$\therefore 64\pi = 16h^2\pi$

$h^2 = 4$

$h = 2, h \geq 0$

$\therefore 2 = -k(0) + C$

$C = 2$

$\therefore h = -kt + 2$

when $h = 0$, $k = 0.001$

$0 = -0.001t + 2$

$t = 2000 \text{ hours}$

Cannot use

$\frac{dV}{dt} = -k(32h\pi)$

since h is
also a variable

MATHEMATICS Extension 1 : Question 6.....

Suggested Solutions

Marks

Marker's Comments

$$(a) \lim_{x \rightarrow 0} \frac{4}{5} \frac{\sin 4x}{4x} + \lim_{x \rightarrow 0} \cos x$$

$$= \frac{4}{5} + 1$$

$$= \frac{9}{5}$$

1 for $\frac{4}{5}$

1 for 1

$$(b) T_{k+1} = \binom{15}{k} \left(\frac{4}{x}\right)^{15-k} (x^2)^k$$

independent of

$$x^{-15+k} x^{2k} = x^0$$

$$-15 + 3k = 0$$

$$3k = 15$$

$$\underline{k = 5}$$

①

$$\binom{15}{5} 4^{10} = 3148873728$$

1

should have
calculated
answer, not
left in
unexpanded
form

MATHEMATICS Extension 1 : Question.....6

Suggested Solutions

Marks

Marker's Comments

(6) Prove true for $n=1$

$$1^3 + 2 \times 1 = 3 \text{ which is divisible by } 3$$

Therefore true for $n=1$

Assume true for $n=k$

$$k^3 + 2k = 3M \quad \begin{matrix} M \in \mathbb{Z}^+ \\ k \in \mathbb{Z}^+ \end{matrix}$$

Prove true for $n=k+1$

$$(k+1)^3 + 2(k+1)$$

$$= k^3 + 3k^2 + 3k + 1 + 2k + 2$$

$$= k^3 + \cancel{2k} + 3k^2 + 3k + 3.$$

by assumption

$$= 3M + 3k^2 + 3k + 3$$

$$= 3(M + k^2 + k + 1)$$

$$= 3I \text{ where } I \text{ is an integer}$$

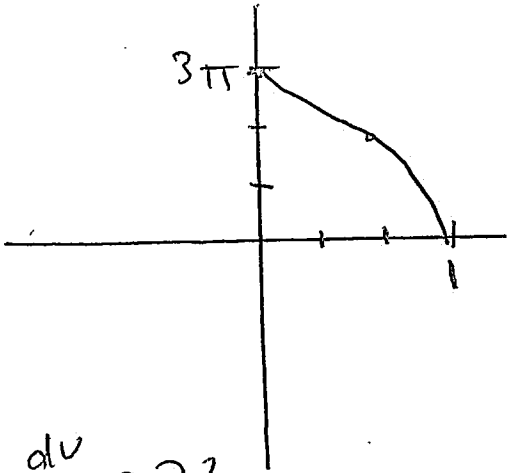
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Needed to state integer

MATHEMATICS Extension 1 : Question.....

MATHEMATICS Extension 1 : Question.....		
Suggested Solutions	Marks	Marker's Comments
d) $f(x) = 3\cos^{-1}(2x-1)$ $-1 \leq 2x-1 \leq 1$ $0 \leq 2x \leq 2$ $0 \leq x \leq 1$ domain $0 \leq y \leq 3\pi$ range 		2 correct facts 1 mark all correct 2 marks there is no horizontal part at 1-flexion.
(e) $\frac{dv}{dt} = 72$ $S = 4\pi r^2 \quad \frac{ds}{dr} = 8\pi r$ $V = \frac{4}{3}\pi r^3 \quad \frac{dv}{dr} = 4\pi r^2$ $\frac{ds}{dt} = \frac{ds}{dr} \cdot \frac{dr}{dv} \cdot \frac{dv}{dt}$ $= 8\pi r \times \frac{1}{4\pi r^2} \times 72$ $= \frac{2}{r} \times 72 \quad r=12$ $= 12$ The rate of increase $12 \text{ m}^2/\text{s}$.	1 Students who substituted V in terms of S needed to use implicit differentiation as both depended on r, so received 0 if they didn't. 1 ignored units	